

RAW SOAP PRODUCTION MINI PROJECT REPORT

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Introduction

Soap has been used for generations, dating back to the Roman empires to the Celts and the Europeans. Soap is one of the oldest chemical processes, yet it remains one of the most essential products in our everyday lives due to sanitation, hygiene and personal care needs. Soap plays a critical role when it comes to public health. For example, simply washing your hands for 30 seconds with hot water and soap can help reduce the transmission of covid-19. It can also be used in multiple other sectors of industries such as textiles and pharmaceuticals.

Now, what is soap? Soap is a substance that when dissolved in water can remove dirt and bacteria from several different surfaces including skin, clothes, and surfaces that are touched daily. The use and production of soap may seem like a very simple process, but it is far more complex than that. The process is further complexified by the addition of fragrances, specific textures, and other add-ins, and there are also variations between liquid and solid soaps. However, the process of making soap typically starts with the creation of soap noodles: small pellets that are a basic combination of the most essential ingredients. Then, these are customized by being melted down, combined with other ingredients, and remolded into a desired final form.

One of the greater challenges when it comes to soap production stems from being sustainable. After use, soap flows into our water system and into the environment which raises concerns about ecological impact. Not only that, but the packages that the soap comes in (whether it is in plastic bottles, cardboard boxes or wrappers) must be designed to be biodegradable or recyclable. The industry faces the task of being able to make soap more environmentally friendly while also maintaining high quality and meeting customers' needs.

Market Analysis

The global soap market continues to grow steadily due to the increase in population and the increasing emphasis on hygiene. Demand surged during the Covid-19 pandemic, and it has remained high as consumers prioritized handwashing and personal care. Several essential raw materials needed—coconut oil, palm oil, and sodium hydroxide—have prices and availability which depend on seasonal variations and global market fluctuations. These variances directly affect the production cost and overall supply stability.

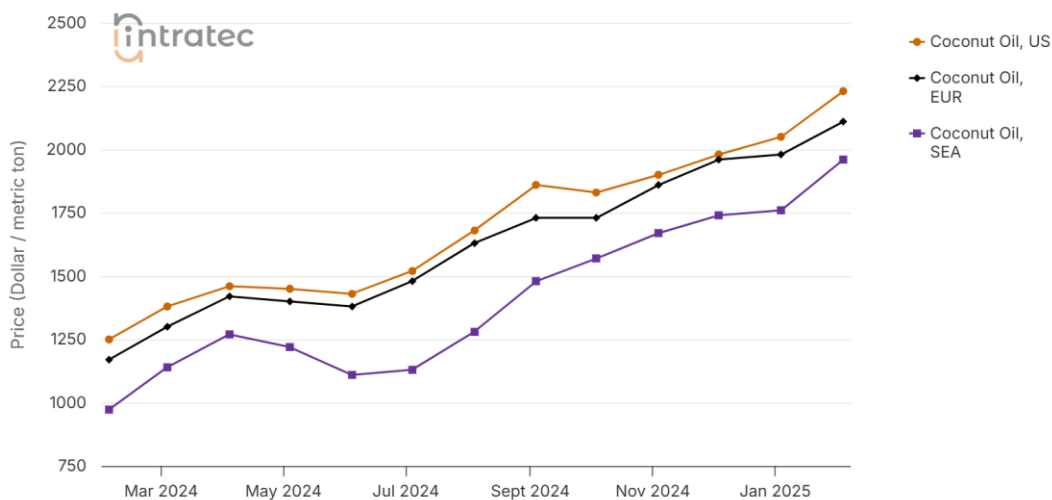


Figure 1: The increase of cost for Tallow

As of 2025, the global soap market is valued at \$50.68 billion and is projected to reach \$66.5 billion in 2032. This considers all types of soaps, including Liquid, bar, foam, scented soaps. The Asia-Pacific region dominates the market, accounting for 38.88% of the global share in 2024. Largely due to the rising population and nationwide hygiene campaigns in both India and China. Bath and body soaps are the most sought after due to their moisturizing and fragrance enhancing properties, while liquid soap is the fastest growing segment. Market preferences also vary by country: For instance, the U.S. market emphasizes organic and antibacterial soap, whereas Japan has stronger demand for eco-friendly packaging and premium liquid soap.

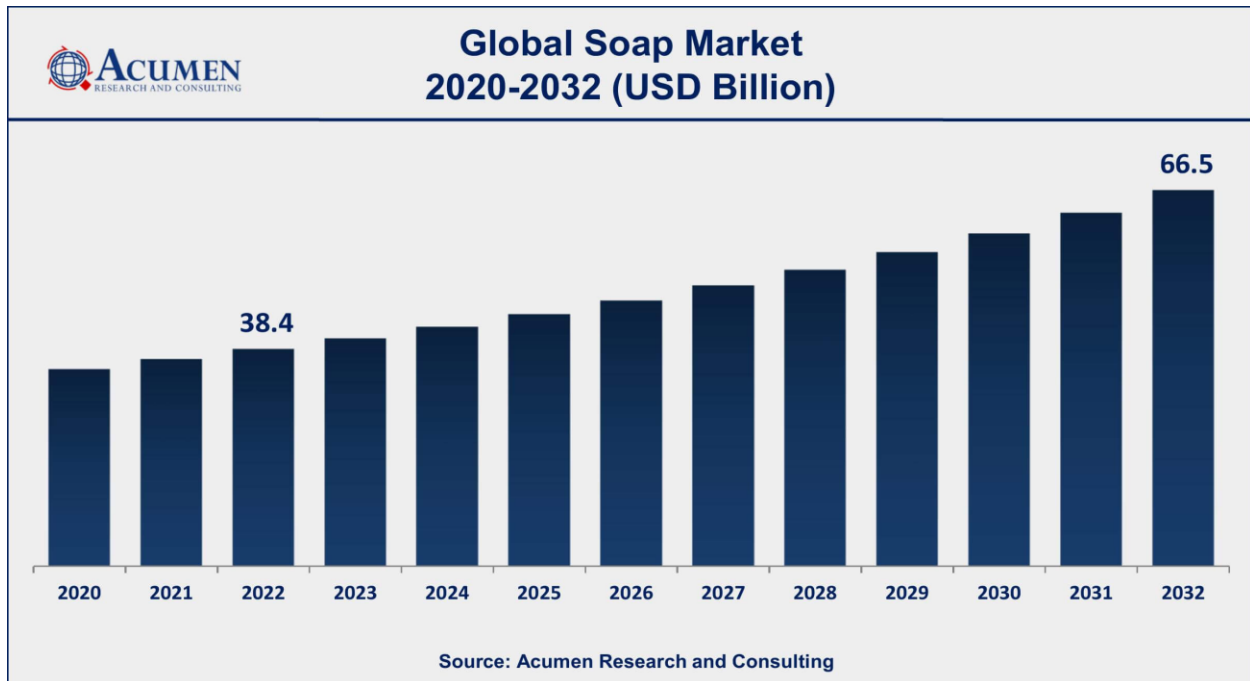


Figure 2: Global soap Market

Several corporations play a role in the soap industry, including Procter and Gamble, Colgate-Palmolive, and Johnson & Johnson. These companies continuously innovate in product formulation and marketing to meet evolving customer expectations by investing in sustainability, product quality, and region-specific demand. They maintain a strong foothold in the global market.

Additionally, a common byproduct of soap production is glycerine, which is highly valuable as it has many uses in other industries. The global glycerine market value at USD 3.79 billion in 2023 and is projected to grow to 4.42 billion by 2032. In the United States the market is expected to grow to approximately 470.13 million dollars, driven by the demand in personal care, pharmaceuticals, and food processing in personal care products it serves as a moisturizer. The compound is produced through saponification.

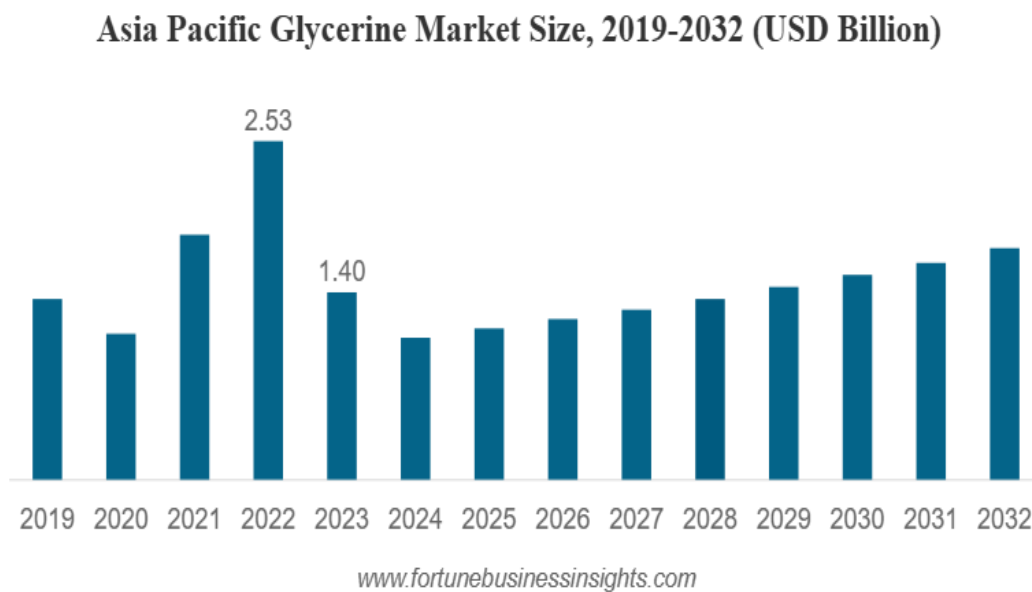
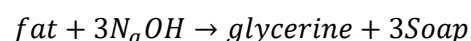


Figure 3: Changes in Asia Pacific Glycerine Market Size over Time

General Process Description

Soaps are primarily made up of two materials: fat and sodium hydroxide. The four common steps to soap making are the following saponification, where animal fat, coconut oil, and sodium hydroxide are mixed and heated. The next step is Glycerin removal, where the glycerin is separated from the soap using salt water. The third step is soap purification, where the remaining sodium hydroxide is neutralized with a weak acid. And finally, the final step is adding preservatives, color, and scents.



Colgate-Palmolive Process

The main process used in soap production is the Colgate-Palmolive process, developed by Binacchi & Co. This continuous method operates with two primary streams. The first stream feeds raw materials into the reactor at an 80:20 tallow-to-coconut oil ratio, with flow rates of 525.9 kg/hr of coconut oil, 131 kg/hr of tallow, and 101 kg/hr of a 50 percent NaOH solution. The second stream introduces a counterflow of lye, which hydrates the soap and facilitates glycerine removal. When the soap and lye mix, they form a wet soap phase. This mixture enters a separator, where the lye settles at the bottom and is sent to a recovery unit, while the soap rises for further processing.

At this stage, the soap still contains residual glycerine. Additional lye is introduced in a washing column to enhance mixing and extract more glycerine. The soap exits at the top of the column, and the lye is pumped away from the bottom for reuse. Remaining NaOH in the soap is neutralized using a weak acid—typically coconut oil fatty acids—and preservatives are added as needed. The final step involves drying the soap by heating it to 125 degrees Celsius under a vacuum pressure of 5.3 kilopascals. As the soap cools to 45 degrees Celsius, it solidifies on the chamber walls, forming the final product.

The Lever Rexona Process

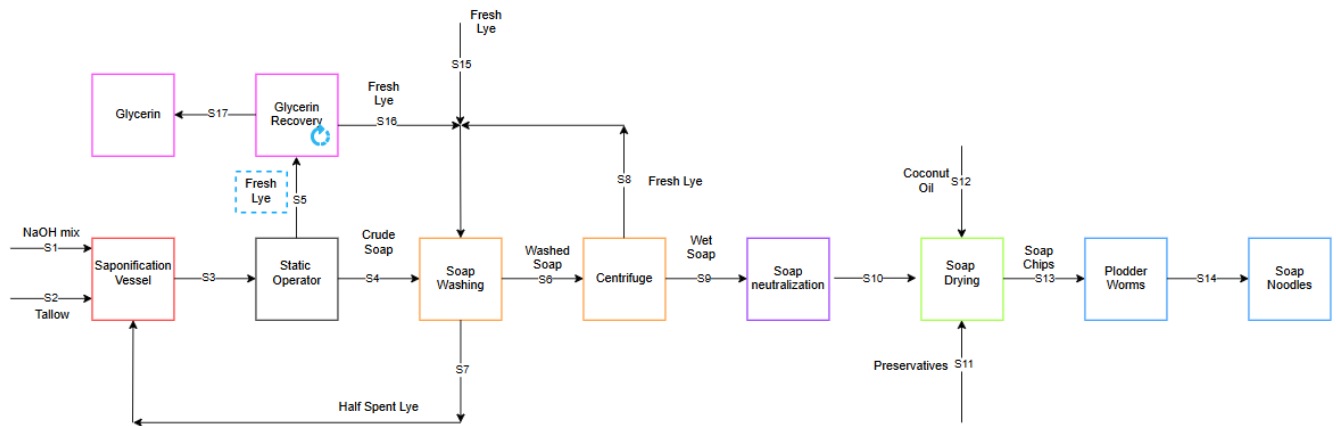
The Lever Rexona process is similar to the Colgate-Palmolive method in that it uses a blend of tallow and coconut oil, which are saponified and dried to produce soap. Unlike the continuous Colgate process, Lever Rexona operates in batch mode. Raw materials are heated together to initiate saponification, converting fats into soap and glycerin. After the reaction is complete, the mixture is washed with lye and brine to separate the phases. The soap rises to the surface, while glycerin and spent lye settle at the bottom. The soap is then sent for further processing, and the lye is recovered for glycerin extraction. Finally, the soap is dried under vacuum conditions to reduce moisture content and produce the final product.

Spray-Drying Process

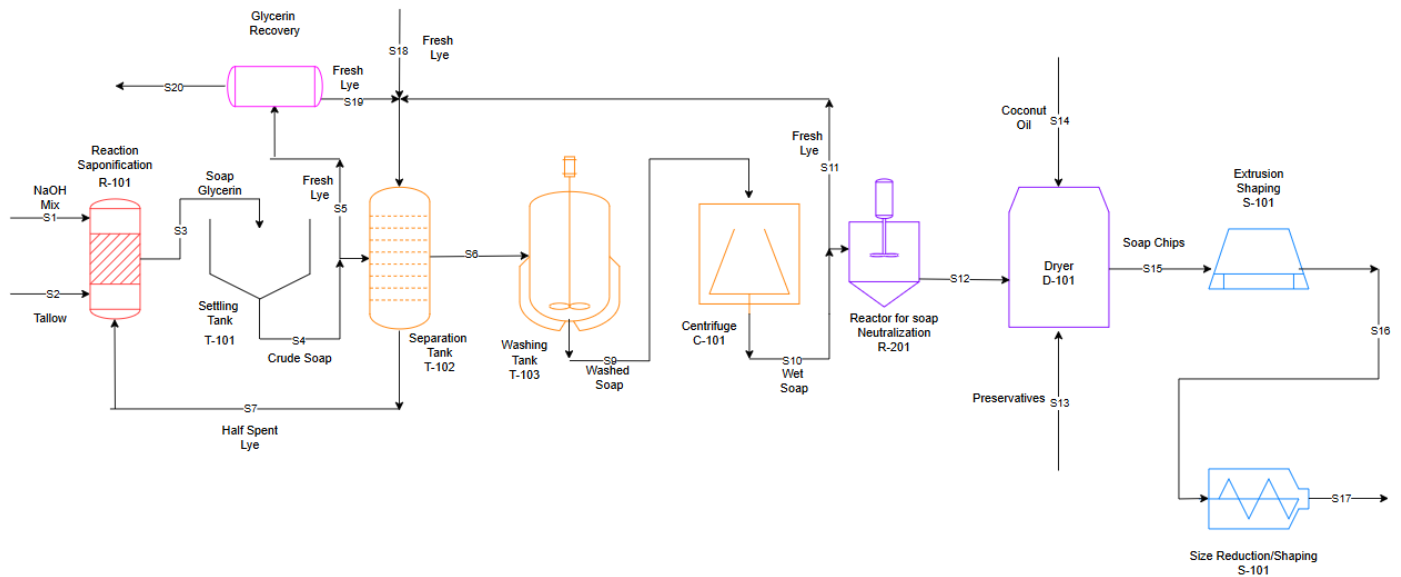
The modern Spray-Drying soapmaking process is also a continuous process in which natural fats are hydrolyzed and then neutralized with lye to produce soap. The initial steps of this process are very similar to the Colgate-Palmolive process, but after the initial soap solution is made, it is pumped into a spray-drying tower. A nozzle or spinning disks are used to atomize the solution into fine droplets and a countercurrent of hot air rapidly removes the water from the soap. The leftover solids fall to the bottom of the tower in flakes, which can then be further processed.

The advantage of this process is that the resulting chips of soaps are easily transportable and can be turned into a variety of different end products, and the process still produces glycerin, a valuable by-product. However, the cost of the spray towers is much higher than the cost of equipment for other soap making processes.

Block Flow Diagram Of Colgate-Palmolive Process



Process Flow Diagram of Colgate-Palmolive Process



Rough material balance

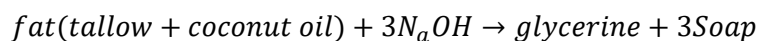
Comparing both processes, it's clear that the Colgate-Palmolive process is better for soap production, so we will make a general material balance around this process, assuming we are making roughly 4,000,000 lbmol/year of product (soap).

COST Materials								
Raw Material	Cost \$/kg	g/mol	lb/mol	Cost \$/lb	Cost \$/mol	Cost \$	Cost/mol of soap	Cost \$/yr soap
Conut oil	2.8	650	1.43	1.27	1.82	0.36	0.373121597	\$ 1,492,486.39
Tallow	1.01	875	1.925	0.46	0.88	0.71		
NaoH	0.42	40	0.088	0.19	0.017	0.05		
						1.12		
Plant size lbmol/year	4000000							
Operatin hours	8000							
Operating days	333.3							
Year	0.913							

The table above shows the cost analysis of how much money in materials would be spent to produce 4,000,000 lbmol/yr of soap using the Colgate-Palmolive process, and as we can see it would cost roughly \$1,492,486 a year to produce 4 million pounds of soap a year.

Glycerine \$/lb	Glycerine lb/mol	\$/mol	\$	Revenue of glycerin	Cost after selling glycerine
3.44	0.203	0.6969	0.6969	929249.49	563236.90

We are also producing glycerin as a byproduct of the soap production process, meaning we can sell this to make this process more cost-efficient, and by selling the glycerin, our cost in materials is reduced to \$547,106, assuming if we sell all the glycerin.



Design Basis:

Plant Size: The plant's size should be chosen based on the number of unit operations needed to meet the yearly demand of soap/detergent production. We decided to work with 4,000,000 lbmol/year of annual soap production using a continuous process.

Raw Materials (raw soap):

Raw materials	Cost \$/lb
Tallow	0.46
Coconut oil	1.27
NaOH	0.19
Water (utility cost)	\$0.00097

Location: We've selected Guildford County in the state of North Carolina as the site for our soap manufacturing plant. This location is a strong choice for soap production due to its access to coastal ports for coconut oil imports, proximity to meat processing facilities for tallow, and a well-developed transportation network for distributing products to major cities on the East Coast. This location has the necessary infrastructure for this process as it is a large logistics hub and chemical cluster. Additionally, North Carolina offers competitive property taxes averaging around 0.63 percent, a corporate income tax rate of 2.25 percent, and an average sales tax of 7 percent. These figures compare favorably to the national average corporate income tax rate of 6.25 percent, making North Carolina a cost-effective and logistically sound choice for our facility.

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